MILITARY SPECIFICATION

POWDER, PROPELLENT
(FOR AMMUNITION, 20MM(AA) GUN)

1. SCOPE

1.1 Scope.--This specification covers propellent powder for use in loading 20mm ammunition for the 20mm/70 Caliber Anti-Aircraft gun.

1.2 Classification.--The powder shall be of one type: single base (nitrocellulose).

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, and drawings, and all specifications listed or referred to in the applicable drawings of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

JAN-D-98	--Diphenylamine.
JAN-G-155	--Graphite (for Use in Ammunition).
JAN-P-193	--Potassium Sulfate (for Ordnance Use).
JAN-E-199	--Ether, Diethyl.
JAN-D-204	--Dinitrotoluene (for Use in Explosives).
JAN-P-528	--Powder, Propellent, M-12
JAN-N-244	--Nitrocellulose (for Use in Explosives).
JAN-A-463	--Alcohol, Ethyl (for Ordnance Use).
MIL-G-2550	--General Specifications for Ammunition Except Small Arms Ammunition.

NAVY DEPARTMENT

General Specification for Inspection of Material.

OS 1906	--Function and Ballistics of Powder, Propellent, for Ammunition 20mm (AA) Gun-40mm (AA) Gun.

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STANDARDS

MILITARY

MIL-STD-129 --Marking of Shipments.

DRAWINGS

LD 328475 --Smokeless Powder Packing Box Mk 7 and Mods.
LD 39249

FB 22274 --Gage, Airspace, Small Arms Ammunition.

(Copies of specifications, standards, and drawings required by contractores in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.--The following publications, of the issue in effect on date of invitation for bids, unless otherwise stated, form a part of this specification:

INTERSTATE COMMERCE COMMISSION REGULATIONS:

Regulations for Transportation of Explosives and Other Dangerous Articles, Etc.

(Information as to availability of these Regulations may be obtained from the Interstate Commerce Commission, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Material.

3.1.1. Raw materials.--The raw materials used in the manufacture of the propellent powder shall conform to the corresponding specifications listed in 2.1, and in addition, the following materials shall be of the type, grade, or class specified hereafter:

3.1.1.1 Nitrocellulose.--Nitrocellulose of type I or II, grade C, of Specification JAN-N-244, made from cotton cellulose shall be used. Nitrocellulose made from wood cellulose shall not be used, unless prior approval of the procuring agency has been obtained. Nitrocellulose recovered from scrap powder shall not be used without prior approval of the procuring agency.

3.1.1.2 Ethyl alcohol.--Grade 2 of Specification JAN-A-463.

3.1.1.3 Graphite.--Grade III or IV of Specification JAN-C-155.

3.1.1.4 Diethyl ether.--Grade A, or Grade C of Specification JAN-E-199.

3.1.1.5 Potassium sulfate.--Grade A of Specification JAN-P-193.

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3.1.2 Rework material.

3.1.2.1 Normal scrap colloid from mixers, blocking presses, extruding presses, cutting machines, and other units shall be collected frequently into covered receptacles, and kept soft with solvents. This colloid may be used in the mixers, either by itself, or mixed with fresh nitrocellulose. It shall be further softened, if necessary, in a macerator.

3.1.2.2 Hard colloid or hard rework from any operation, of the same composition as the material in process, or base grain accumulating between solvent recovery and coating operation, if clean and not damaged by heat or sunlight, may also be softened with solvents in a macerator, and used with fresh nitrocellulose in mixing. The quantity shall be such as to produce a uniform colloid, but shall usually be not over 10 percent of the mixer charge.

3.1.3 Nominal formula of the powder.--The powder shall conform to the corresponding nominal formula, in parts by weight inclusive of coating agent; specified in table I.

TABLE I.--Nominal formula of the powder

CONSTITUENT	
Nitrocellulose	100.00
Potassium sulfate added	1.00
Diphenylamine added	1.00
Dinitrotoluene added	5-12%

The powder shall be coated with dinitrotoluene in amounts necessary to produce the desired characteristics, and shall be graphite glazed.

3.2 Limits on additive chemical constituents (except nitrocellulose).--Limits shall be held, as determined by analysis, on the additive chemical constituents of the powder, in parts by weight, as specified in table II.

3.3 Form.--Cylindrical grains with single perforation.

3.4 External moisture.--1.00 ± 0.25 per cent.

3.5 Total volatiles.--2.35 per cent, maximum.

TABLE II.--Limits on the additive chemical constituents of the powder

ADDITIVE CONSTITUENT	Parts by weight	
	Maximum	Minimum
Potassium sulfate	1.00	.50
Diphenylamine	1.00	.70
Graphite	0.30	.10
Dinitrotoluene	12.00	5.00

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3.6 Residual solvent.--1.10 per cent, maximum.

3.7 Hygroscopicity (per cent moisture absorbed at 30°C., between exposure conditions of 20 per cent and 90 per cent relative humidity).--1.50 per cent, maximum.

3.8 Dust and foreign material.--0.075 per cent, maximum.

3.9 Stability.--When subjected to the 134.5°C. stability test, the powder shall not cause complete fading of the methyl violet test paper to a salmon pink color in less than 45 minutes. Explosion shall not occur in less than 5 hours.

3.10 Loading characteristics.--The powder shall load satisfactorily in all loading machines of the types currently in use.

3.10.1 The bulk density of the powder shall be such that when the powder charge established to develop the required velocity is machine loaded into a cartridge case, and the appropriate bullet is seated to the proper depth in the charged case, the linear compression of the powder charge shall not exceed 0.10 inch.

3.10.2 The extreme variation in weight among 50 consecutive charges of powder thrown from a loading machine of appropriate type using the determined weight of charge established in the velocity firing, shall not exceed 5 grains.

3.11 Function.--The function requirements of this propellant are listed in OS 1906.

3.12 Ballistic requirements.--The ballistic requirements of this propellant are outlined in OS 1906.

3.13 Manufacturing process.--The manufacturing process for the powder shall be approved by the procuring agency, in writing, before the manufacturer proceeds with production. In case any change in the process of manufacture appears to be questionable, the contracting officer shall bring the matter to the attention of the Chief, Bureau of Ordnance, and notify the contractor of this action, in writing, within 24 hours.

3.13.1 With every lot of powder submitted for acceptance the manufacturer shall furnish, on official blanks, 8 copies of a description sheet giving complete history of its manufacture.

3.13.2 Dehydration of nitrocellulose.--The nitrocellulose shall be dehydrated with ethyl alcohol, to remove all water. At least 1.10 pounds of alcohol for every pound of dry nitrocellulose in the press charge shall be pumped through the block, including the residual alcohol left in the block. Excess alcohol shall be removed by pressure, leaving no more than is required for mixing.

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3.13.3 Mixing.--Using suitable mechanical mixers, the blocks of dehydrated nitrocellulose shall be mixed with the necessary amounts of solvent and other ingredients. The amount of solvent used is left to the discretion of the manufacturer. However, the amount of solvent and the mixing time shall be sufficient to give thorough colloiding and distribution of all ingredients. Further colloiding, if necessary, shall be done in a macerator.

3.13.4 Pressing.--The charge from the mixer or macerator shall be blocked for sufficient time, and at a pressure which has been previously determined to give thorough consolidation. The colloid shall then be strained through screens in a "macaroni" press in order to promote more thorough colloiding and to remove uncolloided nitrocellulose, and foreign particles such as pieces of metal and wood. The colloid shall then be blocked and pressed through screens and dies to form strands. The strands shall be granulated, by passing through cutting machines, into grains of such dimensions that, after proper drying, shall have the desired final dimensions and perforations. The screens used shall be such as to reduce to a minimum the amount of uncolloided nitrocellulose in the finished grain. Unless found free of foreign particles, the material left on the screens shall not be reworked.

Note:--Single perforated powders, granulated on vertical presses, may be made without the use of the "macaroni" press, provided the vertical presses contain screening and reforming equipment.

3.13.5 Drying.--The powder shall be subjected to a solvent recovery process at a temperature not exceeding 60°C. If water-drying is used, the temperature shall not exceed 65°C. If air-drying is used, the temperature shall not exceed 55°C. The rate at which the temperature is raised to these limits is left to the discretion of the manufacturer, but it must be such as to prevent distortion of the powder grains.

3.13.6 Coating.--The application of an approved compound to the powder grains, in quantities necessary to give the required ballistic characteristics, shall be performed under conditions of controlled temperature and at the rate of increase in temperature prescribed by the manufacturer.

Note:--Subsequent failure of the powder in machine loading will occur if this operation is not performed according to the prescribed procedure.

3.13.7 Glazing.--The powder shall be glazed with graphite in such manner, and for such length of time, as to insure uniform and complete coverage of the individual grains.

3.13.8 Sieving and cleaning.--The powder shall be sieved and cleaned to remove doublets, clusters, tailings, short cuts, splinters, chips, dust, loose graphite, and foreign materials.

3.13.9 Blending.--The powder shall be blended uniformly in lots of prescribed size. Prior to or during the blending operation, moisture may be added, if necessary in such a way as to insure its uniform distribution, and in such quantity as to meet the requirements for moisture content of the finished powder specified in 3.4.

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3.13.10 Finish.--The propellant powder shall have a clean, polished appearance, shall be uniform in granulation, and the grains shall have such surface characteristics that when it is machine-loaded the powder shall flow and charge uniformly.

3.14 Workmanship.--The finished powder shall be a homogeneous colloid, as free as possible of uncolloided nitrocellulose. Material going through processes shall be kept in covered containers when required, and shall also be protected from direct sunlight, dirt, acids, or contamination. Material found to be harmfully discolored or contaminated shall be removed from processing and destroyed. Contaminated material shall include powder which has been made according to a wrong formula, or which is found to contain material not conforming to the applicable specifications or not otherwise specifically authorized.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.--Unless otherwise specified, a lot shall consist of not less than 25,000 or more than 60,000 pounds of powder of the same class.

4.1.1 If a lot of powder is retreated or redried, it shall constitute a new lot. In such cases the lot number shall be suffixed with the letter "A" to indicate retreatment, and the lot shall be subjected to a complete acceptance test.

4.2 Sampling.--The inspector shall select at random 10 boxes from the lot, and from each selected box he shall remove samples of the sizes specified in table III.

Chemical Sample	Ballistic Sample*	Surveillance Sample	Loading Sample
Ounces	Pound	Pound	Pounds
8	1	1	1½

* Two (2) pounds when lot is the first manufactured under this specification or when directed by the Bureau of Ordnance.

4.2.1 Sampling shall be done with care so as not to handle the powder grains with bare hands, or allow the opened boxes to become contaminated. The chemical and ballistic samples shall be taken directly from the packing boxes, poured into clean bottles, and these at once closed with rubber stoppers. The label on each bottle shall then be marked to show the number of the box from which the sample was taken, in addition to the following information: lot number, type of powder, date of packing, name of manufacturer, plant, and total weight of the packed lot. Each of the 10 bottles containing the 8-ounce samples from the different powder boxes shall be marked "Chemical," while the 1-pound sample bottles shall be marked "Ballistic." Each bottle shall then be sealed with a Government seal. The surveillance samples shall be shipped to Naval Powder Factory, Indian Head, Maryland, or other designated point, for surveillance test. The "Loading" sample shall be used in testing for machine loading characteristics. When shipment of the powder samples is made to a different facility by common carrier, preparation for shipment shall be in accordance with Interstate Commerce Commission Regulations for Transportation of Explosives and other Dangerous Articles, etc.

Note:--These regulations do not permit shipment of powder in glass bottles.

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4.3 Inspection.-- Unless otherwise specified, inspection shall be made at the place of manufacture, and shall be in accordance with Specification MIL-G-2550. Powder which cannot be tested at source of origin shall be forwarded to such facility as may be designated by the procuring agency for the desired testing. The official samples described in 4.2, shall be taken promptly, following the submission of the lot for acceptance.

4.3.1 Inspection procedures.--For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.4 Test procedures.

4.4.1 Chemical composition and physical tests.--The tests shall be conducted in accordance with procedures approved by the Bureau of Ordnance (see 6.3.4) to determine compliance with the requirements of 3.1.3, 3.2, 3.3, 3.4, 3.5, 3.6, 3.7, 3.8 and 3.9.

4.4.2 Loading characteristics.

4.4.2.1 Airspace or compression of powder charge.--Determine the bulk density of powder charge by machine loading 10 cartridge cases, using calibration cases, with the established weight of charge. Do not insert the projectile. Measure the linear distance (either positive or negative, as the case may be) with an airspace gage, as shown on drawing FB22274, between the top of the powder column and the base of the bullet, at the seating position that the bullet would occupy when the cartridge is assembled to the overall length required by the applicable drawing. A negative reading indicates compression, and shall not exceed the limit specified in 3.10.1.

4.4.2.2 Machine loading.--The determined weight of charge shall be dropped from charging blocks, or from plate loaders, into cartridge cases of the type for which the powder is intended. Each charge so dropped shall be verified on a sensitive balance under conditions which will prevent the powder from losing or gaining moisture. The extreme variation in individual charges among the 50 cases so loaded shall not exceed the limit specified in 3.10.2.

4.4.3 Function and ballistics.

4.4.3.1 Procedure.-- All ballistic work, including the use of reference powder, shall be in accordance with the procedures outlined in OS 1906, a sample of reference powder will be supplied by the Bureau of Ordnance for use in function and ballistic tests.

4.4.3.1.1 Powder blend.--Equal amounts of powder shall be taken from each bottle of the official ballistic samples representing different powder boxes, and blended together in a hermetically tight container fitted with a rubber stopper. This blend shall be used to establish the weight of powder charge required to develop the specified velocity and to hand assemble cartridges with the established weight of charge for acceptance firing to obtain the average corrected velocity and recoil.

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4.5 Retest and rejection.--Should a lot of powder fail to pass one or more of the tests specified, the inspector shall notify the manufacturer, and upon the request of the manufacturer shall repeat the failing test or tests. At the time the repeated test or tests are made, the same test or tests shall be made with the "reference powder," and the results obtained shall be used as the basis of comparison for judging the characteristics of the lot powder. Should the "reference powder" fail to give results complying with the specifications, the Government inspector shall investigate the conditions of testing, as well as the "reference powder," in order to determine and to remove the cause of failure before proceeding with the repeated test or tests of the lot of powder for acceptance. If a lot of powder fails to pass the required retest or retests, as described above, it shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packing.--The powder shall be packed in a Packing Box Mark 7 and Mods. All containers before packing shall be tested for air-tightness by the manufacturer and only air-tight containers shall be used. The containers shall withstand without leakage, an internal pressure of 3 pounds per square inch for a period of 1 minute.

5.2 Marking.--Marking shall be in accordance with Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc. Each container shall be marked in accordance with Standard MIL-STD-129, in addition to being marked to show the purchase order number, place and date packed, lot number, nomenclature, class of powder, and the particular ammunition for which the contained powder is intended.

6 NOTES

6.1 Ordering data.--Procurement documents should specify the following:

1. Title, number and date of this specification.

6.2 Although normally furnished in accordance with the formulas, granulations, processes, and desired ballistic properties specified herein, powder deviating in these respects may be procured under this specification, when such specific deviations are authorized in the contract or order.

6.3 Miscellaneous notes.

6.3.1 Surveillance.--Surveillance tests at 65.5°C. and 82.2°C. will be conducted at the Naval Powder Factory, Indian Head, Maryland, for informational purposes.

6.3.2 Acceptance specifications of small arms ammunition require that 90 per cent of the cartridges trace when fired under prescribed conditions. For the purpose of the present specification, satisfactory initiation of igniter and tracer mixtures shall be considered to have been achieved by the powder undergoing acceptance if acceptable trace characteristics can be developed consistently by appropriate ammunition containing such powder, when tested for trace in accordance with the applicable ammunition acceptance specification.

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6.3.3 For purposes of continuing procurement, after a powder has been initially evaluated and found to be acceptable, the tests for determining compliance with the requirements of 3.7, 3.10.1, 3.10.2, 3.11, or of any of them, may be waived upon authorization of the procuring agency.

6.3.4 A specification entitled, "Propellent Powders, General Specification for Methods of Inspection, Sampling and Testing" is in process of preparation. At such time as this specification is officially released, applicable chemical and physical tests will be made in accordance with its provisions. In the meantime, such tests should be made in accordance with procedures in JAN-P-528.

6.3.5 At the time the chemical and ballistic samples are taken by the Government inspector, the manufacturer may, upon authority of the contracting officer, take for his own use samples of the powder in similar quantities and in like manner. These samples may be used by the manufacturer for making ballistic or chemical tests, or both, for his own information.

6.3.6 Copies of assessments for 20mm reference powder may be obtained from the Chief, Bureau of Ordnance, Department of the Navy, Washington 25, D.C.

Notice:--When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:

Navy-Bureau of Ordnance

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